UNIVERSITI SAINS MALAYSIA

School of Electrical and Electronics Engineering
Universiti Sains Malaysia, Engineering Campus

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EEM 441

Control & Instrumentation Laboratory

Experiment 2

DC MOTOR / PID CONTROL — POSITION

Date : ______________________________

Group No. : ______________________________

Group Members : ______________________________ Matrix # __________

: ______________________________ Matrix # __________

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Experiment 2

DC Motor / PID Control — Position

Objectives

1. To study open-loop position control of a DC motor.
2. To study feedback (closed-loop) control using P, PI, PD, and PID controllers.
3. To implement position control of a DC motor using a potentiometer and a resistance-to-voltage converter.

Equipment / Software Required

- Controller kit
- Cathode ray oscilloscope
- BNC connectors with leads
- Multimeter

Procedure

(a) Position control without feedback

1. Connect *Output B* of the motor plant to Channel 2 of the oscilloscope.
2. Connect *Adjust A* of the motor plant to *Input B* (position control) of the plant and to Channel 1 of the oscilloscope.
3. Turn on the power switch.
4. Slowly turn potentiometer-A clockwise (0 to $+50$) and anti-clockwise (0 to -50). Observe and describe the motor operation. Can you control the angular position of the motor?
5. Turn off the motor by disconnecting *Input B*.

Figure 2.1: DC Motor / PID Control trainer.

(b) Closed-loop position control

(i) Position control with proportional feedback control

The motion of a motor can be converted into rotational position with the help of a suitable gearbox. The electrical signal, proportional to (angular) position, is obtained with the help of a circular potentiometer of good accuracy.

This signal is fed back to the summer to extract an error signal, which is fed to the controller to obtain optimum performance. Different controllers may be tried for this case, to study them: use the PID Controller daughterboard shown in Figure 2.2, selecting the desired stage(s) via its DIP switch.

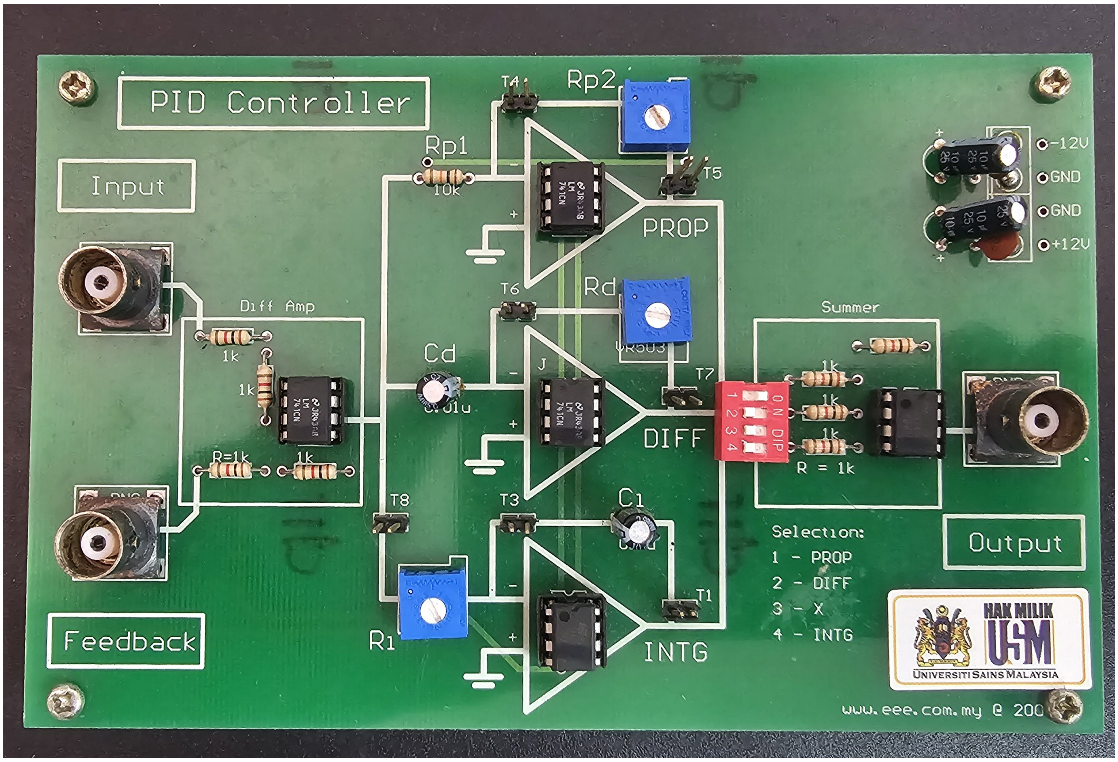

Figure 2.2: PID Control module.

1. Select the P-controller by setting PROP on the controller (DIP switch 1) to ON; all other switches OFF.
2. Connect *Adjust B* of the plant to the *input* of the PID controller.
3. Connect *Output B* of the plant to *Feedback* of the controller.
4. Connect the controller *Output* to *Input B* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. $+180$).
6. Adjust trimmer R_{p2} (varying K_p) to obtain an optimum response (minimum steady-state error).
7. Turn potentiometer-B to vary the position of the motor. Can the position be controlled in this case?
8. Record the input and output voltages for various positions in Table 2.1.

Table 2.1: Position control response.

Position	V_{in} (Channel I)	V_{out} (Channel II)	Error ($V_{out} - V_{in}$)	Remarks
$+270$				
$+240$				
$+210$				
$+180$				
$+150$				
$+120$				
$+90$				
$+60$				
$+30$				

(ii) Position control with feedback and PI-controller

1. Select the PI-controller by setting PROP and INTG to ON.
2. Connect *Adjust B* of the plant to the *input* of the PID controller.
3. Connect *Output B* of the plant to *Feedback* of the controller.
4. Connect the controller *Output* to *Input B* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. $+180$).
6. Slowly vary the respective trimmers R_{p2} and R_i to obtain an optimum output response.
7. Turn potentiometer-B to vary the position of the motor. Record the input and output voltages for various positions.

(iii) Position control with PD-controller

1. Select the PD-controller by setting PROP and DIFF to ON; all other switches OFF.
2. Connect *Adjust B* of the plant to the *input* of the PID controller.
3. Connect *Output B* of the plant to *Feedback* of the controller.
4. Connect the controller *Output* to *Input B* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. $+180$).
6. Slowly vary the respective trimmers R_{p2} and R_d to obtain an optimum output response.
7. Turn potentiometer-B to vary the position of the motor. Record the input and output voltages for various positions.

(iv) Closed-loop control of position of motor using PID controller

1. Select the PID-controller by setting PROP, INTG, and DIFF to ON.
2. Connect *Adjust B* of the plant to the *input* of the PID controller.
3. Connect *Output B* of the plant to *Feedback* of the controller.
4. Connect the controller *Output* to *Input B* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. $+180$).
6. Adjust the value of trimmers R_{p2}, R_i, and R_d to obtain an optimum response (minimum steady-state error, quick operation, without overshoot or oscillations).
7. Turn potentiometer-B to vary the position of the motor.
8. Record the input and output voltages for various positions in Table 2.2.

Table 2.2: Closed-loop PID position response.

Position	V_{in} (Channel I)	V_{out} (Channel II)	Error ($V_{out} - V_{in}$)	Remarks
$+270$				
$+240$				
$+210$				
$+180$				
$+150$				
$+120$				
$+90$				
$+60$				
$+30$				

Report

1. Briefly explain the working of circuits used for the conversion of resistance to voltage.
2. Describe in detail the applications of position control.